

Mehdi Dadfarnia

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Education

University of Maryland, College Park. *College Park, MD.*

Ph.D. Mechanical Engineering, 2025.

Advisor: Jeffrey W. Herrmann. Thesis: *Approaches to Evaluate Condition Monitoring-Based Technologies for Manufacturing Maintenance and Risk Management.*

University of Maryland, College Park. *College Park, MD.*

M.S. Systems Engineering, 2014.

Advisor: John S. Baras. Thesis: *Energy Harvesting Microgenerators for Body Sensor Networks.*

University of Maryland, College Park. *College Park, MD.*

B.S. Electrical Engineering, 2012.

Employment

Research Engineer, Communications Technology Laboratory, National Institute of Standards & Technology.
October 2021 - current

Research Engineer, Engineering Laboratory, National Institute of Standards & Technology.
March 2015 - October 2021

Engineer Trainee, Information Technology Laboratory, National Institute of Standards & Technology.
January 2013 - March 2015

Graduate Research Assistant, Institute for Systems Research, University of Maryland, College Park.
January 2013 - May 2014

Engineer Trainee, Engineering Laboratory, National Institute of Standards & Technology.
June 2011 - January 2013

Honors

United States Department of Commerce Bronze Medal Award; awarded for developing an open-source manufacturing simulator that enables users to plan production and compare AI-based and traditional maintenance policies, January 2025.

Institute for Systems Research's Outstanding Systems Engineering Undergraduate Award, University of Maryland; awarded for contributions to requirements analysis modeling that enabled comparative study of ambulance safety standards, December 2013.

Publications

Peer-reviewed journal articles

Dadfarnia, Mehdi, Michael E. Sharp, and Jeffrey W. Herrmann. "Comprehensive evaluations of condition monitoring-based technologies in industrial maintenance: A systematic review." *Journal of Manufacturing Systems* 82 (2025): 449-477.

Sharp, Michael, Mehdi Dadfarnia, Timothy Sprock, and Douglas Thomas. "Procedural guide for system-level impact evaluation of industrial artificial intelligence-driven technologies: Application to risk-based investment analysis for condition monitoring systems in manufacturing." *Journal of manufacturing science and engineering* 144, no. 7 (2022): 071008.

Bock, Conrad, Raphael Barbau, Ion Matei, and Mehdi Dadfarnia. "An extension of the systems modeling language for physical interaction and signal flow simulation." *Systems Engineering* 20, no. 5 (2017): 395-431.

Peer-reviewed conference articles

Dadfarnia, Mehdi, Michael E. Sharp, Serghei Drozdov, and Jeffrey W. Herrmann. "A simulation-based approach to assess condition monitoring-enabled maintenance in manufacturing." In proceedings of the 2023 7th International Conference on System Reliability and Safety (ICSRS): Bologna, Italy. 2023.

Dadfarnia, Mehdi, and Michael Sharp. "Key elements to contextualize ai-driven condition monitoring systems towards their risk-based evaluation." In proceedings of the 2022 5th International Conference on Artificial Intelligence for Industries (IEEE AI4I): Laguna Hills, CA, USA. 2022.

Dadfarnia, Mehdi, Michael Sharp, and Timothy Sprock. "Understanding and evaluating naive diagnostics algorithms applicable in multistage manufacturing from a risk management perspective." In proceedings of the 2020 International Manufacturing Science and Engineering Conference, Vol. 84263, p. V002T07A042, (ASME MSEC): Virtual. 2020.

Dadfarnia, Mehdi, and Raphael Barbau. "Platform-Independent Debugging of Physical Interaction and Signal Flow Models." In proceedings of the 2019 IEEE International Systems Conference (IEEE SysCon): Orlando, FL, USA. 2019.

Dadfarnia, Mehdi, Conrad Bock, and Raphael Barbau. "An improved method of physical interaction and signal flow modeling for systems engineering." In proceedings of the 2016 Conference on Systems Engineering Research (CSER): Hunstville, AL, USA. 2016.

Dadfarnia, Mehdi, Kamran Sayrafian, Paul Mitcheson, and John S. Baras. "Maximizing output power of a CFPG micro energy-harvester for wearable medical sensors." In proceedings of the 4th International Conference on Wireless Mobile Communication and Healthcare-Transforming Healthcare Through Innovations in Mobile and Wireless Technologies (MOBIHEALTH): Athens, Greece. 2014.

Dadfarnia, Mehdi, Y. Tina Lee, Deogratias Kibira, and Allison Barnard Feeney. "Requirements analysis for safer ambulance patient compartments." In proceedings of the 2013 Conference on Systems Engineering Research (CSER): Atlanta, GA, USA. 2013.

Kibira, Deogratias, Y. Tina Lee, and Mehdi Dadfarnia. "Modeling for optimal ambulance patient compartment layout." In proceedings of the 2012 Spring Simulation Multiconference (SpringSim): Orlando, FL, USA. 2012.

Technical reports

Dadfarnia, Mehdi, and Michael Sharp. "Panel Summary Report: Risk Management & Industrial Artificial Intelligence, INFORMS Annual Meeting, 2021 (Virtual)." NISTIR-8445, National Institute of Standards and Technology (2022).

Barbau, Raphael, Conrad Bock, and Mehdi Dadfarnia. "Translator from Extended SysML to Physical Interaction and Signal Flow Simulation Platforms, Version 1.1." Journal of Research of the National Institute of Standards and Technology 126 (2021).

Barbau, Raphael, Conrad Bock, and Mehdi Dadfarnia. "Translator from extended SysML to physical interaction and signal flow simulation platforms." Journal of Research of the National Institute of Standards and Technology 124 (2019).

Kibira, Deogratias, Y. Tina Lee, and Mehdi Dadfarnia. "Enhancing performance and safety in ambulances through improved design standards." NIST Technical Note 1741, National Institute of Standards and Technology (2012).

Dadfarnia, Mehdi, Y. Tina Lee, and Deogratias Kibira. "A bibliography of ambulance patient compartments and related issues." NISTIR-7835, National Institute of Standards and Technology (2011).

Presentations and posters (excluding conference presentations)

Technical

Modeling and Simulation of Manufacturing Systems. University of Virginia, guest lecture talk for MAE6592: Modeling of Manufacturing Processes (graduate course). Charlottesville, VA, USA. Fall 2022, Spring 2024, &

Spring 2026.

Analytical and Simulation-based Comprehensive Evaluation Approaches for the Condition Monitoring of Industrial Assets. INFORMS Annual Meeting. Atlanta, GA, USA. October 2025.

Tutorial on Discrete-Event Manufacturing Simulation Using Open-Source SimPROCESD Environment. Annual Modeling and Simulation Conference. Washington, DC, USA. May 2024.

Simulating Manufacturing, Maintenance, and Condition Monitoring. INFORMS Annual Meeting. Phoenix, AZ, USA. October 2023.

A Literature Review of Approaches to Evaluate the Operational Risks and Benefits of Condition Monitoring. INFORMS Annual Meeting. Indianapolis, IN, USA. October 2022.

Key Elements of a Risk Management Process for an Asset under Condition Monitoring. INFORMS Annual Meeting. Anaheim, CA, USA. October 2021.

Integrated and/or Executable Systems Modeling Approaches with SysML. University of Maryland, guest lecture talk for ENSE622: Model-Based Systems Engineering (graduate course). College Park, MD, USA. Spring 2019, Spring 2020, & Spring 2021.

Risk and Investment Analysis for Condition Monitoring Systems in Manufacturing. University of Wisconsin-Madison, Internet of Things Systems Research Center, UW IoT Webinar Series. Virtual. January 2021.

Risk-Based Evaluation of Diagnostic Algorithms for Multi-stage Manufacturing. IEEE International Conference on Big Data. Virtual poster. December 2020.

Evaluating Condition Monitoring Algorithms Applicable to Manufacturing, from a Risk Management & Investment Analysis Perspective. INFORMS Annual Meeting. Virtual. November 2020.

Platform-Independent Integration of SysML with 1-D Simulation. NAFEMS Model-Based Engineering conference. Virtual. October 2019.

Professional development

Beginning the Writing Process. National Institute of Standards & Technology, Summer Series for SURF Students. Gaithersburg, MD, USA. July 2022.

Student Supervision

Benjamin Winig. NIST Summer Undergraduate Research Fellowship. *2023 & 2024*

Kevin Song. NIST Summer Undergraduate Research Fellowship (supervised with Sergei Drozdov). *2024*

Ovidio Castillo. NIST Summer Undergraduate Research Fellowship. *2022*

Professional activities

Conference organization

Panel Organizer, NIST Artificial Intelligence for Manufacturing Workshop, 2026

Program committees

IEEE International Conference on Big Data (BIG-DATA)

Winter Simulation Conference (WSC)

Journal reviews

Journal of Simulation

Conference reviews

IISE Annual Conference & Expo (IISE-ANNUAL)

Winter Simulation Conference (WSC)

IEEE International Conference on Big Data (BIG-DATA)

Service

Institutional service

Member of NIST Post-docs and Early-career Association of Researchers (PEAR) Leadership Board, 2021-2026

Weekly Writing Group Organizer for NIST-PEAR, 2019-2023

University departmental service

President of the Institute for Systems Research Student Organization at the University of Maryland, 2013-2014

Skills

Engineering expertise: Discrete-event modeling & simulation, digital twin design, prognostics & health management, applied machine learning, risk analysis, engineering decision-making.

Engineering software: Simulink, SysML (proficient), Modelica, Arena (past experience).

Programming: Python, MATLAB (proficient), Julia, C/C++ (past experience).

Libraries: pandas, NumPy, SciPy, scikit-learn, Matplotlib, SimPy, ManPy (proficient).

Languages: English (native), Persian (proficient).